

1	Course Name:	Mobile Application Development																																																																																																																																																																																																		
	Course Code:	BMIT2073																																																																																																																																																																																																		
	Course Classification:	Elective (core)																																																																																																																																																																																																		
2	Synopsis:	Students would gain an insight of mobile technologies for creating mobile applications. Topics covered include categories of mobile applications, mobile application models, operating platforms, user interface, data storage, and onboard instruments. Students will also be exposed to development tools for creating mobile applications.																																																																																																																																																																																																		
3	Name(s) of Academic Staff:	1	Refer to timetable																																																																																																																																																																																																	
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4	Semester and Year offered:	Year Offered		Semester		Remarks: Refer to timetable																																																																																																																																																																																														
5	Credit Value:	3																																																																																																																																																																																																		
6	Pre-requisite/ co-requisite (if any):	BACS2023 Object-Oriented Programming																																																																																																																																																																																																		
7	Course Learning Outcomes (CLO) 	CLO1	Apply mobile technologies and development methods to solve a given problem. (C3, PLO2)																																																																																																																																																																																																	
		CLO2	Apply mobile development methods to develop a mobile application. (P4, PLO3)																																																																																																																																																																																																	
		CLO3	Present a mobile application to be published in a team. (A2, PLO4)																																																																																																																																																																																																	
8	Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods																																																																																																																																																																																																			
<table border="1"> <thead> <tr> <th rowspan="2">Course Learning Outcomes</th> <th colspan="11">Programme Learning Outcomes (PLO)</th> <th rowspan="2">Teaching Methods</th> <th rowspan="2">Assessment Methods</th> </tr> <tr> <th>PLO 1</th> <th>PLO 2</th> <th>PLO 3</th> <th>PLO 4</th> <th>PLO 5</th> <th>PLO 6</th> <th>PLO 7</th> <th>PLO 8</th> <th>PLO 9</th> <th>PLO 10</th> <th>PLO 11</th> </tr> </thead> <tbody> <tr> <td>CLO1</td> <td></td> <td>v</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>L, P, NF2F</td> <td>Test, Practical Assessment and Examination</td> </tr> <tr> <td>CLO2</td> <td></td> <td></td> <td>v</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>L, P, NF2F</td> <td>Assignment</td> </tr> <tr> <td>CLO3</td> <td></td> <td></td> <td></td> <td>v</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>L, P, NF2F</td> <td>Assignment</td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td rowspan="3">Mapping with MQF Cluster of Learning Outcomes</td> <td></td> <td>C2</td> <td>C3A</td> <td>C3B</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td rowspan="3"></td> <td rowspan="3"></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>													Course Learning Outcomes	Programme Learning Outcomes (PLO)											Teaching Methods	Assessment Methods	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	CLO1		v											L, P, NF2F	Test, Practical Assessment and Examination	CLO2			v										L, P, NF2F	Assignment	CLO3				v									L, P, NF2F	Assignment																																																																												Mapping with MQF Cluster of Learning Outcomes		C2	C3A	C3B																																		
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Indicate the primary causal link between the CLO and PLO by ticking 'v' in the appropriate box. C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C = Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Entrepreneurial Skills, C5 = Ethics & Professionalism																																																																																																																																																																																																				
9	Transferable Skills (if applicable) <i>(Skills learned in the course of study which can be useful and utilized in other settings)</i> <table border="1"> <tr> <td>1</td> <td>Cognitive skills</td> </tr> <tr> <td>2</td> <td>Interpersonal skills</td> </tr> <tr> <td>3</td> <td></td> </tr> <tr> <td colspan="2">Open-ended response (if any)</td> </tr> <tr> <td>4</td> <td></td> </tr> </table>												1	Cognitive skills	2	Interpersonal skills	3		Open-ended response (if any)		4																																																																																																																																																																															
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10	Distribution of Student Learning Time (SLT)																																																																																																																																																																																																			

Note: This SLT calculation is designed for home grown programme only.

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities**										Total SLT
			Face-to-Face (F2F)								NF2F Independent Learning (Asynchronous)		
			Physical				Online/ Technology-mediated (Synchronous)						
L	T	P	O	L	T	P	O						
1	Introduction to Mobile Application Development -Mobile operating systems and ecosystem -Development tools -Mobile devices -Mobile application software architecture and framework	1, 3	2	0	2							2	
2	Mobile Applications Models -Screen and event handling -Linking screens -Application life cycle	1, 2, 3	2		2							2	
3	User Interfaces -UI layouts -UI components -Input events -Menus -Dialogs and notifications	1, 2, 3	8		8							8	
4	Resources and Data Storage -Providing and accessing to resources -Shared preferences -Internal storage -External storage -Database -Network and Background Operation -Mobile App Backend	1, 2, 3	8		8							8	
5	Location-based Services -Maps -Getting location data	1, 3	2		2							2	
6	Specialised Instruments and Devices -Camera -Audio capture -Media playback	1, 3	4		4							4	
7	Mobile Application Packaging and Publication -Preparing for publishing -Publishing on the market -Updating application -Monetising methods	1, 3	2		2							2	
8													
9													
10													
11													
12													
13													
14													
15													
16													
17													
18													
19													
20													
SUB-TOTAL SLT:													84
			Face-to-Face (F2F)								NF2F		

Continous Assesement		%	Physical	Online/ Technology-mediated (Synchronous)	Independent Learning for Assessment (Asynchronous)
1	Test	10	1		2
2	Practical Assessment	25	2		4
3	Assignment	35	2		15
4					
5					
SUB-TOTAL SLT:					26
Final Assesement		%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment (Asynchronous)
			Physical	Online/ Technology-mediated (Synchronous)	
1	Examination	30	2		8
2					
3					
4					
5					
SUB-TOTAL SLT:					10
SLT for Assessment:					36
GRAND TOTAL SLT:					120
A	% SLT for F2F Physical Component: $\frac{[Total\ F2F\ Physical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$				52.50
B	% SLT for Online & Independent Learning Component: $\frac{[(Total\ F2F\ Online + Total\ Independent\ Learning)]}{[(Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning)]} \times 100]$				47.50
C	% SLT for All Practical Component: $[\% F2F\ Physical\ Practical + \% F2F\ Online\ Practical]$				23.33
C1	% SLT for F2F Physical Practical Component $\frac{[Total\ F2F\ Physical\ Practical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$				23.33
C2	% SLT for F2F Online Practical Component $\frac{[Total\ F2F\ Online\ Practical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100]$				0.00

Please tick (v) if this course is **Industrial Training/ Clinical Placement/ Practicum** using 50% of Effective Learning Time (ELT)

Note:

* Indicate the CLO based on the CLO's numbering in Item 8

** For ODL programme: Courses with mandatory practical requirements imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

11	Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)	Mobile application development integrated development environment (IDE). E.g. Android Studio. Computer with mobile device emulators and active Internet connection.
12	References (include required and further readings, and should be the most current)	Main reading list: 1. Apple Developer Documentation. (2025). Apple Developer Documentation. https://developer.apple.com/documentation/ 2. Flutter Documentation. (2025). Flutter Documentation. https://docs.flutter.dev/ 3. Guide to App Architecture. (2025). Android Developers. https://developer.android.com/topic/architecture Additional reading list: 1. Essential Kotlin. (2025). Essential Programming Books. https://www.programming-books.io/essential/kotlin/ 2. Essential iOS. (2025). Essential Programming Books. https://www.programming-books.io/essential/ios/ 3. Essential Dart. (2025). Essential Programming Books. https://www.programming-books.io/essential/dart/
13	Other additional information (if applicable)	
Note: Number of PLO indicated is purely for illustration purposes only and the number is subjected to the curriculum design.		